Data

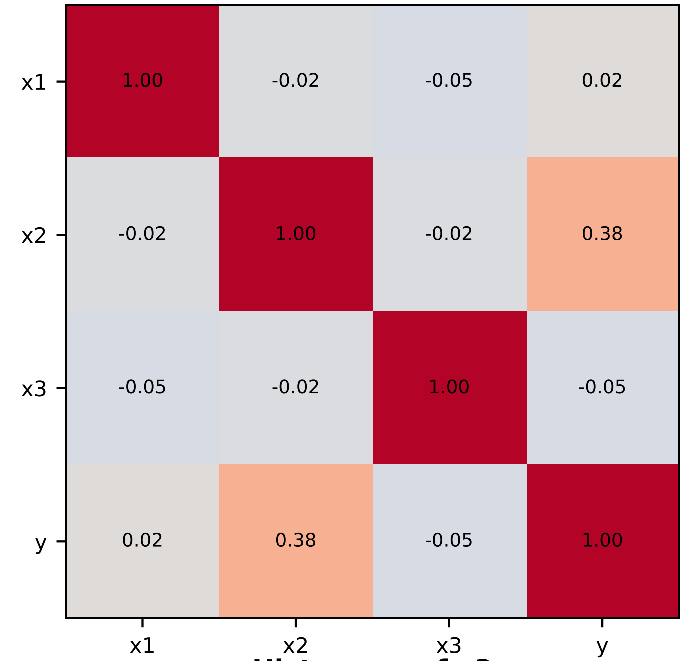
Sample Size & Structure

Training observations: 300
Test observations: 100
Predictors: x1, x2, x3
Response: y (binary)

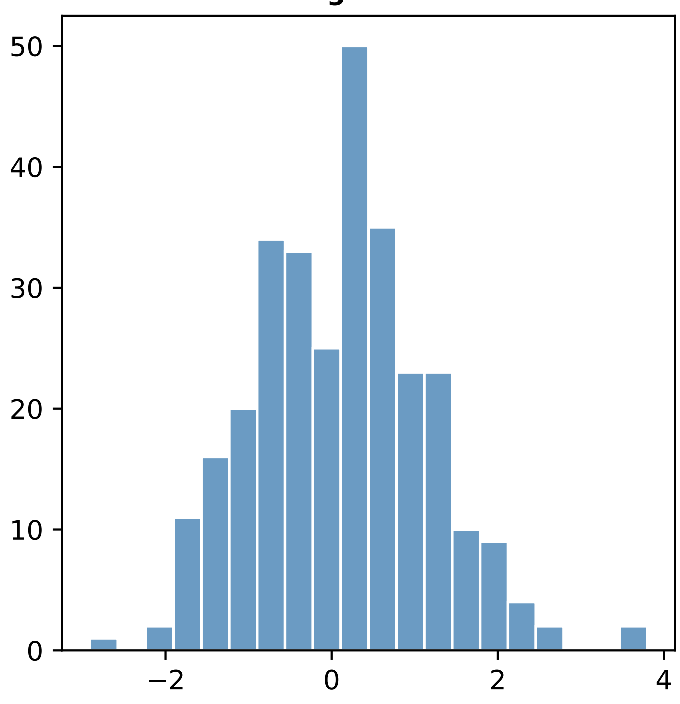
Summary Statistics

	x1	x2	x3	y
mean	0.11	-0.02	-0.06	0.49
std	1.03	0.98	0.99	0.50
min	-2.91	-2.49	-2.71	0.00
max	3.80	3.04	2.70	1.00

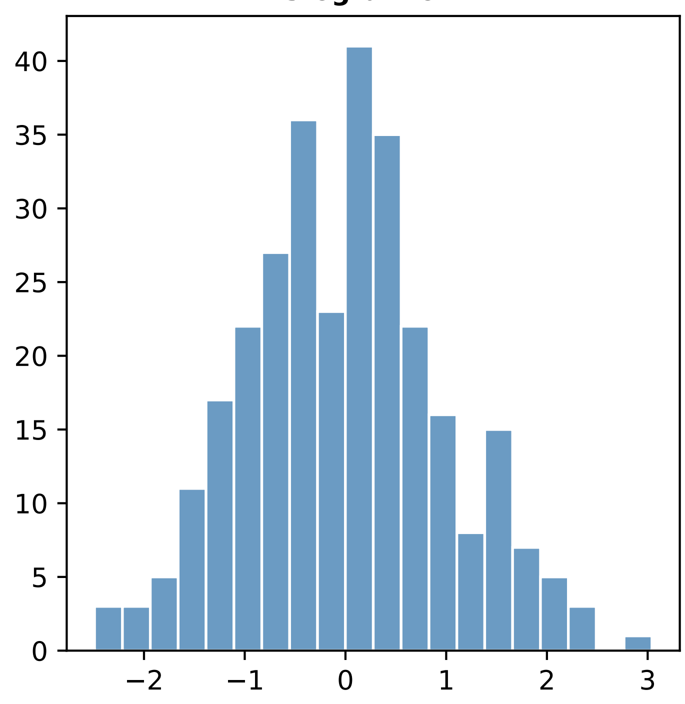
Correlation Heatmap



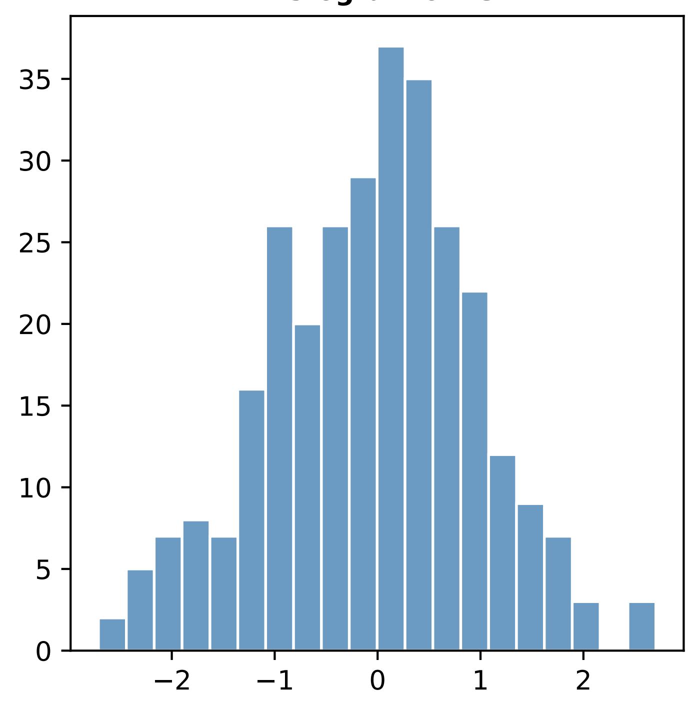
Histogram of x1



Histogram of x2



Histogram of x3



Model, Method and Results

Assumptions

Model Assumptions:
1. Binary Outcome: y is strictly 0 or 1.
2. Independence: observations are i.i.d.
3. Linearity of Log-Odds: Addressed by adding degree-2 polynomial features.
4. No severe multicollinearity.

Method & Libraries

Method & Libraries:
- Logistic Regression (Logit)
- Polynomial Feature Expansion (deg=2)

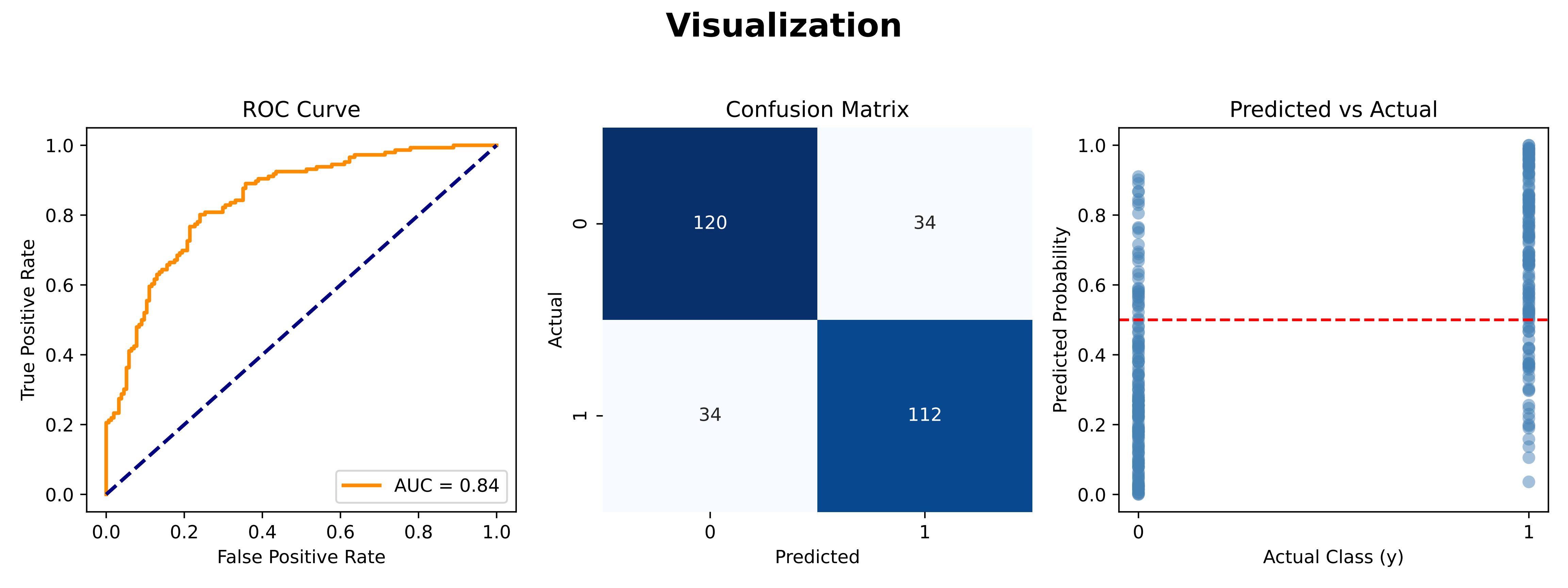
Libraries Used:
- statsmodels.api (Logit modeling)
- scikit-learn (Metrics & preprocessing)
- pandas & numpy (Data handling)
- matplotlib & seaborn (Visualizations)

Formula & Quantities

```
Logistic Regression Formula:
  logit(p) = -0.289
             -0.070*x1
             +1.254*x2
             -0.396*x3
             +0.128*x1^2
             -0.109*x1 x2
             +1.602*x1 x3
             -0.015*x2^2
             +0.062*x2 x3
             +0.195*x3^2
```

Coefficients Summary

Term	Coef	p-value
const	-0.289	0.2072
x1	-0.070	0.6895
x2	1.254	0.0000
x3	-0.396	0.0370
x1^2	0.128	0.3024
x1 x2	-0.109	0.5731
x1 x3	1.602	0.0000
x2^2	-0.015	0.9243
x2 x3	0.062	0.7489
x3^2	0.195	0.1970


Visualization
ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
AUC = 0.84
Confusion Matrix
Actual
Predicted
0
1
120
34
34
112
Predicted vs Actual
Predicted Probability
Actual Class (y)